

Module 03: Perception — How Agents Make Sense of the World

Course: Active Inference 101 | **Unit:** Cognitive Science | **Audience:** First-semester undergraduates

Learning Objectives

1. Explain **predictive processing**: the brain as a prediction machine that constructs perception from expectations.
2. Define **prediction error** and describe how it drives perceptual inference.
3. Analyze examples where perception fails (illusions, hallucinations) to understand the constructive nature of perception.

Introduction

You might think perception is simple: light enters your eyes, and you see the world. But perception is far more active and constructive than that. Your brain doesn't passively receive information — it *predicts* what it should see, hear, and feel, then checks those predictions against incoming sensory data. When predictions and data match, you perceive the world smoothly. When they don't, you get a **prediction error** — and your brain must update.

Key Concepts

1. The Brain as a Prediction Machine

The **predictive processing** framework says the brain is constantly generating predictions about sensory input:

- Before you see a face, your brain predicts what a face looks like based on context
- Before you hear a word, your brain predicts what word comes next based on the sentence so far
- Before you step off a curb, your brain predicts the height of the step

Perception happens when your brain's **top-down predictions** meet **bottom-up sensory data**. What you "see" is actually your brain's best guess about what's out there.

2. Prediction Error and Bayesian Inference

When prediction and data don't match, the brain computes a **prediction error**:

Prediction Error = Actual Sensation – Predicted Sensation

The brain then updates its internal model (its beliefs) to reduce this error. This is a form of **Bayesian inference** — updating beliefs based on new evidence:

- **Prior**: What the brain expected before seeing the data
- **Likelihood**: How well the data fits different hypotheses
- **Posterior**: The updated belief after combining prior and data

You don't need the formal math yet. The key idea: **perception = prior beliefs + sensory evidence**, weighted by how confident the brain is in each.

3. Precision — How Much to Trust Your Senses

Not all prediction errors are treated equally. The brain assigns **precision** (confidence) to different sources of information:

- In a bright room, the brain trusts visual data (high visual precision)
- In a dark room, the brain trusts its expectations more (low visual precision, high prior precision)
- When you're anxious, the brain may over-weight threatening predictions (high precision on threat priors)

Precision is like a volume knob: it controls how loudly prediction errors "speak" to the brain.

4. Illusions as Prediction Errors

Visual illusions reveal the constructive nature of perception:

- **The Hollow Mask Illusion:** A concave face looks convex because your brain's prediction ("faces are convex") overrides the sensory data
- **The McGurk Effect:** Hearing "ba" while watching someone mouth "ga" produces the percept "da" — your brain fuses conflicting audio-visual predictions
- **Change Blindness:** You fail to notice large changes in a scene because your brain predicted continuity

These aren't "bugs" — they're features of a predictive system doing its best with limited data.

5. From Perception to Cognition

Perception in Active Inference is the process of minimizing prediction error by **updating beliefs** (changing the model to fit the world). But there's another option: you can minimize prediction error by **acting** (changing the world to fit the model). Module 05 covers this. First, Module 04 explores how the brain organizes and optimizes its beliefs — cognition.

Summary

Perception is not passive reception but active prediction. The brain generates top-down predictions and compares them with bottom-up sensory data, computing prediction errors that drive belief updating. Precision weighting determines how much the brain trusts different information sources. Illusions reveal this constructive process in action.

Further Reading

- Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3), 181-204.
- Hohwy, J. (2013). *The Predictive Mind*. Oxford University Press.
- Rao, R. P. N. & Ballard, D. H. (1999). Predictive coding in the visual cortex. *Nature Neuroscience*, 2(1), 79-87.